

Characteristics and Performance of Three Low-Cost Superpressure Balloon (Tetroon) Systems

N. J. CHERRY

Physics Dept., University of Canterbury, New Zealand

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ABSTRACT

The characteristics and performance of superpressure balloons in general and tetroons in particular are discussed. Parameters of tetroons such as cross-sectional area, volume and drag coefficients are evaluated. Three balloon systems are discussed: sealed tetroons, valved tetroons and tandem balloons. The ascent rates of the three systems are determined and compared. The effect of a large forced change in height of the superpressure balloon from its equilibrium height is discussed and expressions for the superpressure, stress and volume perturbations are derived.

1. Introduction

In undertaking a lee-wave project it was decided that superpressured, constant volume balloons (SPB's) were the best source of data, there being no suitable aircraft available. SPB's have been used successfully in many lee-wave projects, though sometimes only in a secondary role (e.g., Booker, 1965; Reynolds *et al.*, 1968; Wooldridge and Lester, 1969; Vergeiner and Lilly, 1970). For this project a reasonably fast ascent rate was required so that the balloons would be at the level of interest by the time they crossed the mountains producing the waves. Booker *et al.* (1968) achieved this by using an aircraft as a launching platform. Booker and Cooper (1965) and Wooldridge and Lester used tandem balloon systems which consisted of a tug or lifting balloon (liftoons) which lifted the SPB until they were separated by a fuse near the SPB's equilibrium level. This method proves unsatisfactory in cases such as the present project, when the balloons are usually released in heavy rain. A pressure sensitive device using an aneroid capsule was developed in order to separate the balloons at a pressure level. Later a gas release valve was developed which allowed the lifting gas to escape from the balloon as it ascended but retained a superpressure as the balloon approached its equilibrium level. This allowed the SPB to be filled completely before release, giving maximum lift and a high mean ascent rate.

Tetroons have often been used for constant volume balloon flights. Angell (1963) used them to determine the Lagrangian properties of turbulence. Angell *et al.*, (1966) used tetroons to study low-level air flows in an urban atmosphere. However, very little is found in published literature on the general properties and performance characteristics of tetroons. Here an at-

tempt is made to evaluate most of the parameters required in the basic use of tetroons. Some expressions pertaining to SPB's in general are derived. These are useful in calculating their performance characteristics and in estimating their limitations.

2. Tetroons

Some of the characteristics of tetroons are found in Angell and Pack (1960), Angell (1963), Booker and Cooper (1965), Delver and Booth (1965) and Whannel (1965). Tetroons are used because of their ease of manufacture and their relatively high volume to surface area ratio. A tetroon with sides of length L has an approximate cross-sectional area in the apex up or apex down orientation given by¹

$$A^* = 0.485L^2 \quad (1)$$

compared to $0.433L^2$ for a tetrahedron. The difference of about 12% is due to the bulging of the sides of the tetroon. The displacement volume of an $L=1$ m, 1 mil polyester (Mylar or Melinex) tetroon when filled to have a superpressure of 10 mm of Hg was found to be 0.177 m^3 . Therefore, the volume of a tetroon was taken to be given by

$$V = 0.177L^3 \quad (2)$$

Thus, the area required to enclose 1 m^3 of gas in a tetroon is 5.48 m^2 compared to 4.83 m^2 for a sphere, a difference of only 13.5%. Whannel (1965) used 2-mil polyester tetroons and found that $A^* = 0.493L^2$ and $V = 0.182L^3$. Presumably these are for a superpressure somewhat greater than 10 mm.

¹ A complete list of symbols is given in Appendix A.

TABLE 1. Equilibrium height H_e (km), ascent rate $\bar{v}$ (m sec⁻¹) and time T (min) to ascent to the equilibrium level from sea level for sealed tetrooms of various sizes l (m) with representative loads L (gm).

L		40	60	80	100	Load 120	140	160	180	200
1.2	H_e	6.42	5.46	4.59	3.80	3.00	2.25	1.56	1.92	0.31
	$\bar{v}$	1.20	1.12	1.05	1.00	0.95	0.91	0.87	0.84	0.80
	T	89.6	81.4	72.6	63.4	52.6	41.4	30.0	18.3	6.4
1.4	H_e	8.92	8.22	7.51	6.82	6.17	5.57	5.01	4.48	4.00
	$\bar{v}$	1.43	1.36	1.29	1.23	1.17	1.13	1.10	1.05	1.01
	T	104	101	97.2	92.6	87.6	82.4	77.0	71.4	65.8
1.6	H_e	10.7	10.2	9.60	9.08	8.60	8.14	7.66	7.19	6.73
	$\bar{v}$	1.66	1.58	1.51	1.45	1.40	1.35	1.30	1.25	1.22
	T	108	107	106	104	103	101	98.2	95.1	92.0
1.8	H_e	12.3	11.8	11.3	10.8	10.4	10.0	9.61	9.24	8.89
	$\bar{v}$	1.83	1.77	1.72	1.67	1.61	1.55	1.51	1.47	1.43
	T	111	111	109	108	108	107	106	105	104
2.0	H_e	13.5	13.1	12.7	12.3	11.9	11.6	11.2	10.9	10.6
	$\bar{v}$	1.99	1.94	1.89	1.84	1.79	1.75	1.71	1.68	1.63
	T	113	113	112	111	111	110	109	108	108
2.2	H_e	14.6	14.2	13.9	13.5	13.2	12.9	12.6	12.3	12.0
	$\bar{v}$	2.14	2.09	2.04	2.00	1.95	1.91	1.87	1.83	1.80
	T	114	114	113	113	113	112	112	111	111
2.4	H_e	15.6	15.3	14.9	14.6	14.3	14.0	13.8	13.5	13.2
	$\bar{v}$	2.29	2.24	2.19	2.14	2.10	2.06	1.99	1.98	1.96
	T	114	114	114	114	114	113	113	113	114
2.6	H_e	16.5	16.1	15.9	15.6	15.3	15.1	14.8	14.6	14.3
	$\bar{v}$	2.42	2.24	2.33	2.29	2.35	2.21	2.17	2.14	2.11
	T	113	113	113	113	113	113	113	113	113

The free lift of a balloon, from Archimedes principle, is given by

$$F_L = 0.359(\rho_a - \rho_\theta) \frac{PV}{T} - W, \quad (3)$$

where $\rho_a = 1.293 \text{ kg m}^{-3}$ and $\rho_\theta = 0.0899 \text{ kg m}^{-3}$ for hydrogen and 0.1785 kg m^{-3} for helium. Assuming the 1966 Standard Atmosphere, we have

$$\rho = 0.465 \frac{P}{T}. \quad (4)$$

The equilibrium height of a balloon can be found from tables of the ambient density as a function of height using an expression derived from (3) and (4), since at the equilibrium height the free lift is absorbed as a superpressure. Hence, in (3), $F_L = 0$. Therefore,

$$\rho_e = \frac{1.295W}{(\rho_a - \rho_\theta)V}. \quad (5)$$

A particular equilibrium height is obtained in two main ways:

1) By using a single size of balloon and adding ballast. This adjusts W in (5) until the appropriate value of ρ_e is obtained.

2) By varying the size of the balloon used. This adjusts both W and V .

Table 1 gives some representative values of the equilibrium heights of 1-mil polyester tetrooms of various sizes with various loads.

Lally (1967) showed that the superpressure for a sealed balloon, assuming zero supertemperature and no stretching, is given by

$$p_s = \frac{F_L}{W} p_e. \quad (6)$$

3. Drag coefficients of tetrooms

An important factor in the calculation of the ascent rate of a balloon is its drag coefficient. This is not a constant but varies with Reynolds number and orientation to the air flow. To determine the drag coefficients of a tetroom a model was made by coating a small inflated tetroom with plaster of paris. This gave a shape that was very close to that of a fully inflated tetroom. The model had $L = 0.31 \text{ m}$. The drag coefficients of this model were determined for two orientations, corresponding to ascending with its apex up and with its apex down, by placing it in the wind tunnel of the Department of Mechanical Engineering, University of Canterbury. Air speeds of about 5 to 35 m sec⁻¹ were used, corresponding to Reynolds numbers from about 1×10^5

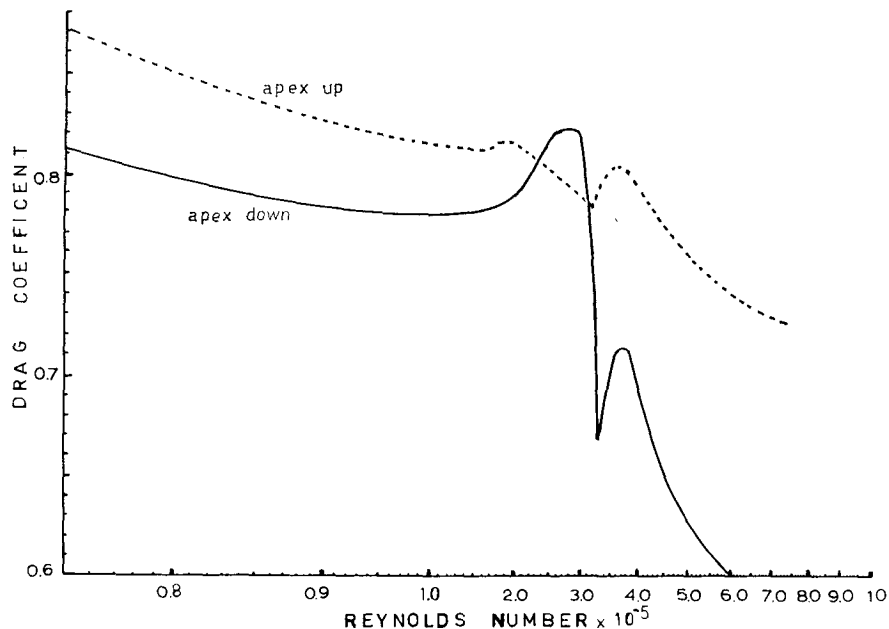


FIG. 1. Drag coefficients of a model of a tetrao ($L=0.31$ m) in two orientations, corresponding to a tetrao ascending with apex up and apex down, as a function of Reynolds number.

to 7×10^5 . The results are shown in Fig. 1. The sudden drop in C_D at a Reynolds number of 3×10^5 is of interest. It corresponds to the "drag crisis" observed for spheres

(Streeter, 1962). The general form of the curves is very similar to that for spheres, which is to be expected, although the drag on a tetrao is somewhat greater.

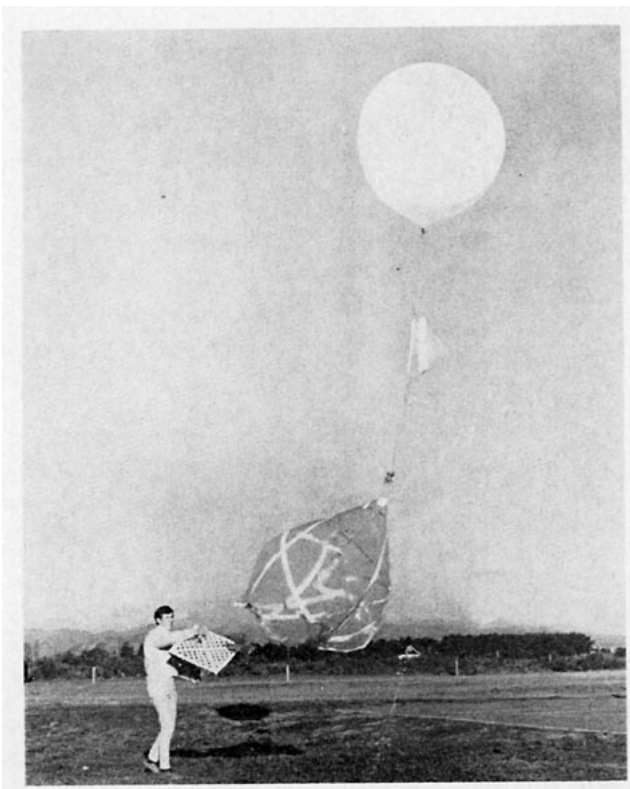


FIG. 2. A tandem balloon system consisting of a 100-gm liftoen with 750 gm of lift, a pressure trigger, a 3-mil, taped polyethylene tetrao ($L=2.1$ m), and a corner radar reflector.

4. Balloon systems

The three main balloon systems used are described in the following paragraphs.

a. Sealed tetraons

A sealed tetraon consisted of a tetraon which was partially filled before release with enough gas to raise the balloon and its load, but not enough to exceed the maximum superpressure allowed by the strength of the balloon film when it reached its equilibrium height.

b. Tandem balloons

A tandem system (Fig. 2) consisted of a sealed tetraon which was towed by a liftoen to a height just below its equilibrium height. Here the balloons were separated by a pressure-operated release mechanism. Such a mechanism is shown in Fig. 3. The expansion of an aneroid capsule was used to withdraw a pin from a spring-loaded rod from which the SPB was supported. The aneroid capsule was positioned so that the pin was withdrawn at a preset pressure height.

c. Valved tetraons

A simple gas release valve involving a spring-loaded ping-pong ball (Fig. 4) allowed the tetraons to be completely filled before release. This generally provided

enough lift to result in a reasonable ascent rate. The ascent rate decreased as the balloons ascended as the lifting gas was lost through the valve to maintain the superpressure at the valve operating pressure. The valve closed as the balloon approached its equilibrium height to retain a superpressure to give the balloon a constant mean equilibrium height. Fig. 5 is a photograph of a 1.4 m radius, 1-mil polyester valved tetraon with a pressure corner reflector.

5. Practical aspects of the balloon systems

Icing is a major cause of the failure of balloons (Vergeiner and Lilly, 1970). If a balloon gets wet as it ascends the water is very likely to freeze and build up a coating of ice. The weight of the ice can easily exceed the free lift of the balloons. It was calculated that it would require a layer of ice only 0.3 mm thick on the top surface of the liftoon alone to absorb 1 kg of lift. V. E. Lally of NCAR Project GHOST suggested the application of a silicon-based aerosol furniture polish to shed water from the balloons in order to reduce the buildup of ice. Balloon failure could also be caused by the failure of the pressure release mechanism to separate the balloons in a tandem system because of icing. This was overcome by coating the release mechanism with a silicon-based oil. A fast ascent rate was also employed to reduce icing by decreasing the time spent ascending through clouds and the freezing layer.

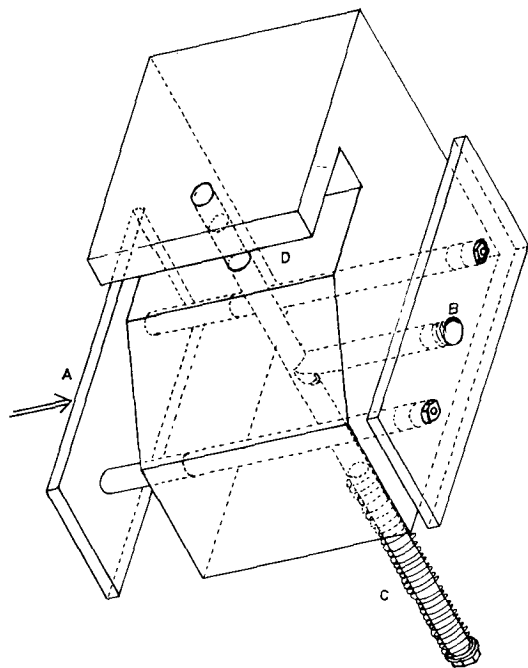


FIG. 3. The trigger mechanism of a pressure-operated release mechanism for use with a tandem system. An aneroid capsule depresses the plate A withdrawing the pin B so that the spring C ejects the rod, dropping the SPB tether from the slot D.

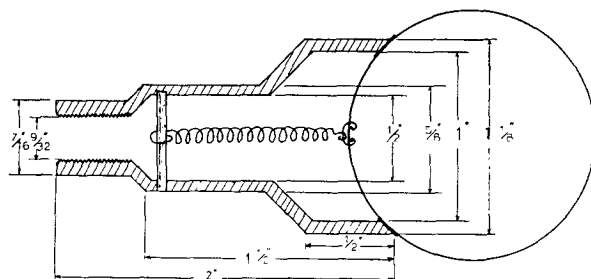


FIG. 4. A cross section through a gas release valve, showing the spring-loaded ping-pong ball. A good seal between the rod and the ball is obtained by a silicon rubber seal around the rim of the rod. The valve operating pressure is set by adjusting the tension of the spring.

A fast ascent rate can be obtained with a tandem system, independent of the size of the tetraon and the strength of the tetraon material. Both of these limit the ascent rates of sealed tetraons and the size of the balloon limits the ascent rates of valved tetraons.

It was found for this project that valved tetraons were the best to work with. They did not require careful weighing off as the sealed balloons did when used on their own or in a tandem system. They were far easier to launch, especially in the open windy conditions often encountered, and less could go wrong which would cause the failure of the balloon flight. One drawback with a valved balloon compared to a sealed balloon, which is discussed below in more detail, is the possibility that in a large amplitude lee wave the pressure change due to the displacement may be great enough to cause

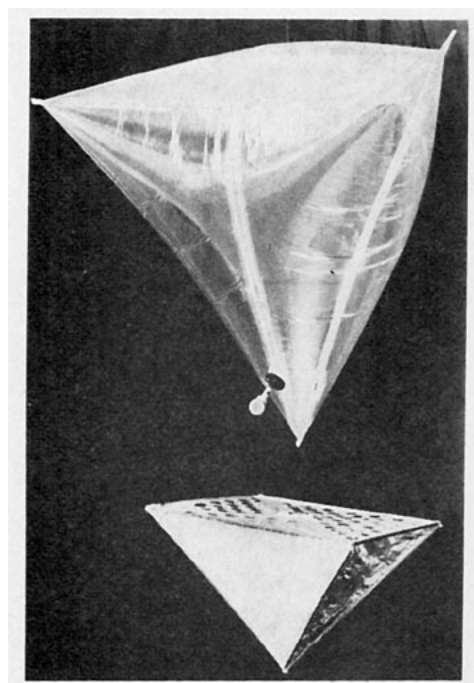


FIG. 5. An $L=1.4$ m, 1-mil polyester valved tetraon with a passive corner reflector.

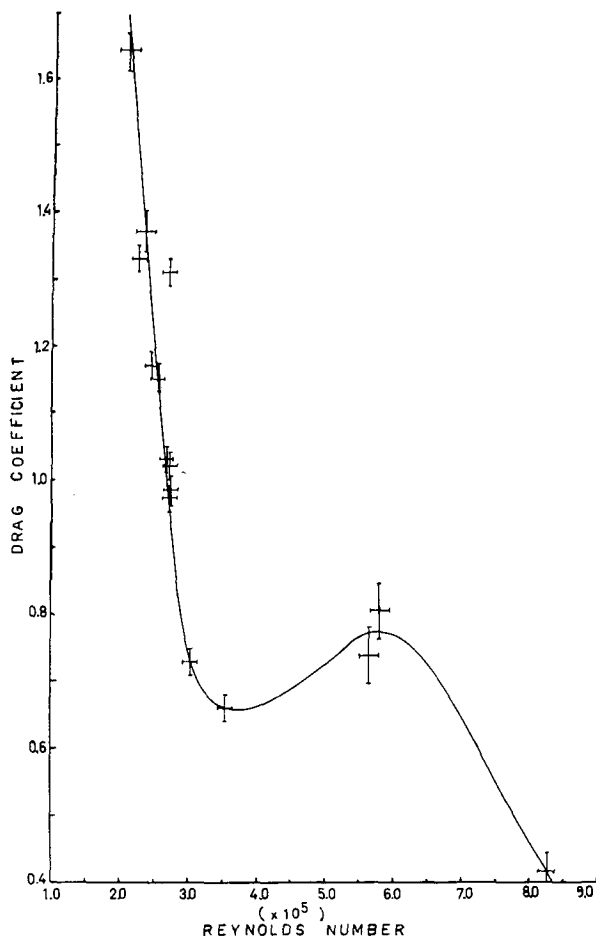


FIG. 6. The drag coefficient of a tandem system as calculated from 17 tandem balloon flights. The effective area was taken as the cross-sectional area of the liftoons at the SPB's equilibrium height. Errors bars indicate the approximate uncertainty of the calculations, mainly due to the uncertainty in the free lift of the liftoons.

the loss of all the lifting gas producing the stabilizing superpressure.

6. Drag coefficients and ascent rates of tandem systems

Representative values of ascent rates of tandem systems were approximately as follows: 3 m sec⁻¹ for 800 gm of lift in a 100-gm liftoon; 4 and 5.3 m sec⁻¹ for 1000 and 2000 gm of lift, respectively, in a 500-gm liftoon; and about 8 m sec⁻¹ for 3000 gm of lift in a 650-gm liftoon.

Because of the turbulent flow past each of the components in a tandem system the drag was not equal to the sum of the drag on each of the components separately. The upper components, mainly the liftoon, tended to shield the lower components. The amount of shielding depended on the ascent rate and the relative sizes of the components. To account for this in calculating the ascent rates of tandem systems, an empirical

curve of the drag coefficient of a tandem system vs Reynolds number was constructed using data from 16 actual tandem balloon flights (Fig. 6). The effective area of a tandem system was taken as the cross-sectional area of the liftoon at the SPB's equilibrium height. The drag of the corner reflectors was taken as the drag of a flat plate of the same area. Since they had an area of 0.274 m², $C_{DA}^* = 0.274$ for the corner reflector.

An indication of the magnitude of the shielding can be gained from the following example: A tandem system consisting of a 500-gm liftoon with 1000 gm of free lift, towing an $L = 1.7$ m tettoon having 20 gm of free lift with a corner reflector attached, had an ascent rate of ~ 4 m sec⁻¹. Considering the drag on the system to be due to the sum of the drag on each of the components, the drag equation will be of the form

$$F_L = \frac{1}{2} \rho v^2 C_{DA}^*, \quad (7)$$

where

$$C_{DA}^* = C_{DL}A_l^* + C_{DL}A_i^* + C_{DL}A_r^*, \quad (8)$$

with l , i and r referring to the tettoon, the liftoon and the radar reflector, respectively. The parachute and pressure trigger are small enough to be neglected. Taking the values of the drag coefficients as given by the empirical curves in Fig. 1 and Streeter (1962), and using $C_{DL}A_r^* = 0.274$, it was found that the system would require a free lift of ~ 1500 gm to attain the ascent rate of 4 m sec⁻¹. Thus, the shielding in this case reduced the total drag by about 33%.

7. Ascent rates of valved balloons

In general the ascent rate of a balloon is governed by an equation of the form of (7). The free lift is given by (3). Equating (3) and (7), rearranging, and noting that F_L in (3) is in kilograms as compared to newtons (N) in (7), we have

$$v = \left\{ \frac{2g \left[0.359(\rho_a - \rho_g) \frac{pV}{T} - W \right]}{C_{DA}^* \rho \times 10^3} \right\}^{\frac{1}{2}}. \quad (9)$$

Using (4) to eliminate P/T , Eq. (9) becomes

$$v = \left\{ \frac{2g \left[0.772(\rho_a - \rho_g) V - \frac{W}{\rho} \right]}{C_{DA}^* \times 10^3} \right\}^{\frac{1}{2}}, \quad (10)$$

where $C_{DA}^* = C_{DL}A_l^* + C_{DL}A_r^*$. [Note: ρ is the ambient density and ρ_a and ρ_g are at STP] For a valved balloon that has been completely filled before release the volume and the effective area are constant. In (10) then, the only variable is the ambient density; this decreases with height as the balloon ascends until at the equilibrium level it reaches the value

$$W/\rho = 0.772(\rho_a - \rho_g)V. \quad (11)$$

When (11) is satisfied the ascent rate is zero. Fig. 7 shows the height vs time curves for valved tetrooms of various sizes made of 1-mil polyester and carrying a load of 180 gm. It can be seen that the maximum time taken by a tetroom of any size to reach its equilibrium level is about 80 min. The calculation assumes negligible shielding of the radar reflector by the tetroom.

8. Ascent rates of sealed balloons

Any gas causing lift must, for a sealed balloon, be compressed at the equilibrium height to produce a superpressure. A relation between the superpressure and the ascent rate can be found by eliminating F_L between (6) and (7), noting again the different units for F_L ; thus, we have

$$p_s = K_B v^2, \quad (12)$$

where

$$K_B = (\frac{1}{2} p_e C_D A^* \rho / g W) \times 10^3,$$

with K_B being a constant for a particular balloon. Values of A^* and ρ both vary as the balloon ascends, A^* increasing and ρ decreasing, but since it has been observed that the ascent rates of sealed balloons are essentially constant it is assumed that the factor $(A^* \rho)$ is constant. Thus, the maximum allowed ascent rate for a particular balloon is

$$v_{\max} = \left(\frac{p_{s\max}}{K_B} \right)^{\frac{1}{2}}. \quad (13)$$

The maximum superpressure that a balloon can contain will be determined by the maximum stress which the balloon film can tolerate. To derive an expression relating the superpressure to the film stress it is noted that the superpressure may be given by (Newman and Searle, 1963)

$$p_s = T_s \left(\frac{1}{r_1} + \frac{1}{r_2} \right). \quad (14)$$

The surface tension may be defined as the force per unit width of cross section. The surface tension divided by the thickness of the film is equal to the film stress. Thus, the film stress (Nm^{-2}) produced by a given superpressure is

$$\sigma = \frac{p_s r_1 r_2}{t(r_1 + r_2)}. \quad (15)$$

Eq. (15) shows that the greatest stress occurs at those points where the balloon film has the least curvature. For a tetroom this occurs at the center of the triangular faces where $r_1 = r_2 \approx L$. For a sphere $r_1 = r_2 = L$. Hence, the maximum stress for a given superpressure is

$$\sigma_m = \left(\frac{2.625 p_s L}{t} \right) \times 10^6. \quad (16)$$

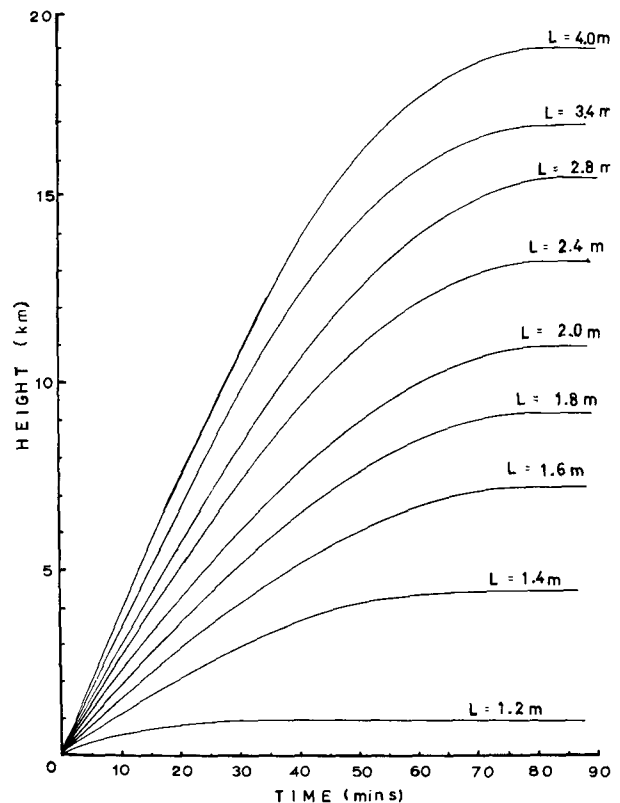


FIG. 7. Height vs time curves for various size 1-mil polyester valved tetrooms carrying a load of 180 gm. Calculations were made using Eqs. (5) and (10) assuming no shielding of the radar reflector by the tetroom.

Therefore, the maximum superpressure that a spherical balloon or a tetroom can stand is given by

$$p_{s\max} = \left(0.381 \sigma_{\max} \frac{t}{L} \right) \times 10^{-6}. \quad (17)$$

For low-density polyethylene, $\sigma_{\max} = 6.9 \times 10^6 \text{ N m}^{-2}$, while for polyester, $\sigma_{\max} = 1.722 \times 10^8 \text{ N m}^{-2}$. Thus, for 1-mil polyester

$$p_{s\max} = \frac{65.6}{L} [\text{mm of Hg}]. \quad (18)$$

For an $L = 1 \text{ m}$, 1-mil polyester tetroom (18) gives a value of 65.6 mm as the maximum superpressure. A destruction test was carried out on a similar balloon. It burst at a superpressure of 62 mm. This is in good agreement with (18). However, for safety, in calculating the allowable free lift we used

$$p_{s\max} = \frac{40}{L} [\text{mm of Hg}]. \quad (19)$$

Similar expressions may be derived for other plastics and other thicknesses of film.

Using (5), (13) and (19), Table 1 was constructed assuming the 1966 Standard Atmosphere. Table 1 shows the equilibrium height, mean ascent rate and approximate time taken to ascend to the equilibrium height from sea level, for various representative loads and tetron sizes.

9. The effect of a large change in height

The response of SPB's to vertical air motions has been studied by Booker and Hirsch (1966). They showed that in regular sinusoidal airflows the balloon followed the motion of the air quite closely with a slightly smaller amplitude and a small phase difference. If a balloon is forced to fly above its equilibrium height, as in large-amplitude lee waves, two major factors need to be considered to see whether the balloon can tolerate such a displacement without being overpressured and bursting:

1) The decrease in atmospheric pressure will cause the differential pressure across the balloon film to increase, increasing the superpressure and the film stress. If this increase in stress is large enough it will cause a sealed balloon to burst or a valved balloon to lose all of the gas producing the superpressure and free lift, thus causing the eventual permanent loss of height.

2) The increase in stress will tend to stretch the balloon film (elastically or plastically depending on the magnitude of the stress), increasing its volume and decreasing the net superpressure.

Expressions for the increase in superpressure, volume and stress due to a forced change in height are derived in Appendix B. They are as follows:

$$\Delta p_s = -\frac{1}{6C} + \frac{\Delta p - p_I}{2} \pm \left[\frac{1}{36C^2} + \frac{(\Delta p + p_I)}{6C} + \frac{(\Delta p - p_I)^2}{4} \right]^{\frac{1}{2}}, \quad (20)$$

$$\Delta \sigma = -\frac{Y}{6} + \frac{CY}{2}(\Delta p - p_I) \pm \left[\frac{Y^2}{36} + \frac{CY^2}{6}(\Delta p + p_I) + \frac{C^2Y^2}{4}(\Delta p - p_I)^2 \right]^{\frac{1}{2}}, \quad (21)$$

or

$$\Delta \sigma = CY\Delta p_s, \quad (21a)$$

and

$$\Delta V = 3CV\Delta p_s. \quad (22)$$

An indication of the order of displacement that SPB's can tolerate is found by considering a 1-mil polyester tetron whose equilibrium height is 5 km ($p_e = 406$ mm, $L = 1.4$ m) with an equilibrium level superpressure of 10 mm. If it is displaced upward through 1 km ($\Delta p = 52$ mm) then from (20), (21a) and (22), $\Delta p_s = 24.4$ mm,

$\Delta \sigma = 8.9 \times 10^7$ N m⁻², and $\Delta V = 0.0342$ m³. The increase in volume reduces the increase in superpressure to less than half the decrease in the atmospheric pressure. However, the increase takes the superpressure of the tetron up to 34.4 mm which is over the safe superpressure, defined by (19), of 28.6 mm for a 1-mil, $L = 1.4$ m tetron. The balloon should not burst since the superpressure does not exceed the maximum allowable superpressure of 47 mm as defined by (18). Thus, a sealed tetron, such as described above, could tolerate a 1-km upward displacement without bursting, although a valved tetron would not tolerate such a displacement as Δp_s exceeds the operating superpressure and too much gas would be lost. On the other hand, if the balloon's initial superpressure was greater than 24.4 mm, a valved balloon would survive a 1-km displacement retaining a small equilibrium level superpressure, but a sealed balloon would have burst. Hence, for lee-wave studies either a sealed balloon at a low superpressure towed to its equilibrium height by a liftoon or a valved balloon at a relatively high valve operating pressure is suitable. It may be noted that a larger balloon or a balloon at a higher level would experience a smaller increase in superpressure for a 1-km upward displacement and a stronger balloon material could stand a greater maximum superpressure.

10. Comparison and conclusions

The tandem system works well in getting the SPB's quickly to their equilibrium heights. The reliability of tandem systems to date has been about 70% in a

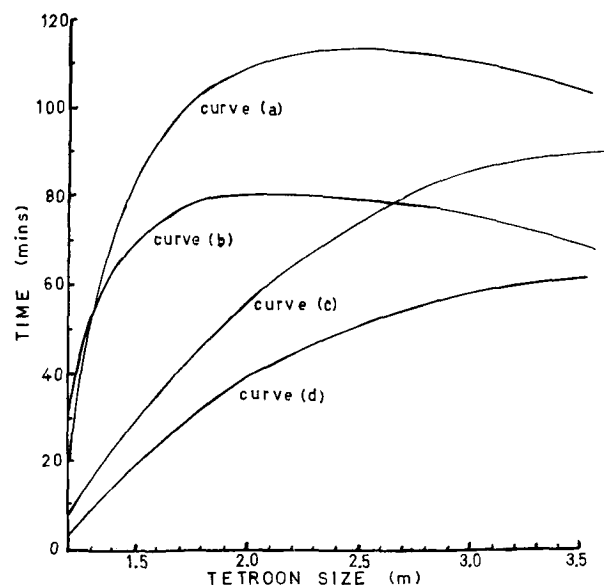


FIG. 8. The time taken by tetrons to reach their equilibrium heights from sea level as a function of tetron size, for the three systems (assuming a load of 180 gm): 1) sealed tetrons, 2) valved tetrons, and 3) a tandem system using a 100-gm liftoon with 700 gm of free lift; and 4) a tandem system using a 500-gm liftoon with 1500 gm of free lift.

sample of 18, with most of the failures being due to icing. Difficulty may be experienced in releasing a tandem system in windy conditions if adequate shelter is not available.

Tetroons make good SPB's if they are made with a strong material such as polyester. Taped polyethylene tetroons and polyester tetroons have been used successfully in tandem systems. The ascent rates of sealed balloons are too small for short-duration flights. They are limited by the small amount of gas initially allowed so that the balloon will not be overstressed at altitude. They may be suitable at low levels (see table 1). The addition of a gas release valve allows the balloon to be completely filled before release. This increases the mean ascent rate considerably. Only one valved balloon has failed to date out of a sample of 20.

To compare the ascent rates of the sealed, tandem and valved systems a 1-mil polyester tetroon was chosen to have a typical load of 180 gm. Ascent rates, and thus ascent times, were calculated for these tetrons for sizes ranging from $L=1.2$ to 3.5 m. The time taken for each of the various sized balloons to reach their equilibrium heights was plotted for 1) sealed tetroons, 2) valved tetroons, 3) a tandem system with a 100-gm liftoons with 700 gm of free lift, and 4) a tandem system with a 500-gm liftoons with 1500 gm of free lift. The results, shown in Fig. 8, can be used when choosing the type of system required for a particular application.

Acknowledgments. I wish to thank I.C.I. of Great Britain for generously supplying me with a large roll of Melinex. Without this the scope of my project would have been severely limited. Also the assistance given by the New Zealand Meteorological Service in supplying personnel and radar equipment has been invaluable.

APPENDIX A

List of Symbols

A^*	effective cross-sectional area of a balloon (m^2)
C	constant for a particular balloon ($\text{m}^3 \text{N}^{-1} \text{mil}^{-1}$)
C_D	drag coefficient
F_L	free lift of a balloon
g	acceleration due to gravity (m sec^{-2})
K_B	constant for a particular balloon ($\text{mm of Hg sec}^2 \text{m}^{-2}$)
L	length of the side of a tetroon or radius of a spherical balloon (m)
P	ambient atmospheric pressure (mm of Hg)
p_e	ambient pressure at the SPB's equilibrium height (mm of Hg)
p_I	internal pressure of a balloon at its equilibrium height (mm of Hg)
p_s	superpressure of a SPB (mm of Hg)
$p_{s\max}$	maximum superpressure a balloon can tolerate (mm of Hg)
Δp	decrease in ambient pressure due to a change in height (mm of Hg)

Δp_s	increase in superpressure due to a forced change in height (mm of Hg)
r_1, r_2	radii of curvature of the balloon surface in two perpendicular planes (m)
ρ	ambient density (kg m^{-3})
ρ_a	density of air at STP (kg m^{-3})
ρ_e	ambient density at the SPB's equilibrium height (kg m^{-3})
ρ_g	density of the lifting gas at STP (kg m^{-3})
σ	balloon film stress (N m^{-2})
$\Delta\sigma$	increase in film stress due to a forced change in height (N m^{-2})
σ_m	maximum stress for a given superpressure (N m^{-2})
$\sigma_{\max}$	ultimate tensile stress of the balloon film (N m^{-2})
t	thickness of the balloon film (mils)
T	ambient temperature ($^{\circ}\text{K}$)
T_s	surface tension of the balloon film (N m^{-1})
v	balloon ascent rate (m sec^{-1})
V	volume of a balloon (m^3)
ΔV	change in volume due to a forced change in height (m^3)
W	weight of balloon and its load (kg)
Y	modulus of elasticity of the balloon film (N m^{-2})

APPENDIX B

Superpressure, Stress and Volume Perturbations due to a Forced Change in Height

The superpressure of a balloon at its equilibrium height is given by

$$p_s = p_I - p_e. \quad (1a)$$

If a balloon is forced to rise so that the ambient pressure drops by Δp , the balloon's volume will increase due to stretching by ΔV and the internal pressure of the balloon will change to p_I . The new superpressure will be

$$p_{sn} = p_i - (p_e - \Delta p). \quad (2a)$$

We then substitute the gas laws

$$p_I V = p_i (V + \Delta V), \quad (3a)$$

and Eq. (1a) in (2a) to obtain

$$p_{sn} = p_s + \Delta p - [\Delta V / (V + \Delta V)] p_I. \quad (4a)$$

Thus, the increase in superpressure, $p_{sn} - p_s$, is

$$\Delta p_s = \Delta p - [\Delta V / (V + \Delta V)] p_I. \quad (5a)$$

If the volume is fixed, i.e., $\Delta V = 0$, then (5a) gives $\Delta p_s = \Delta p$ as expected. If, however, there is an increase in volume then the resulting increase in superpressure is less than Δp . The volume is of the form $V = KL^3$, so if the side of the tetroon is increased by ΔL , the increased volume will be

$$V + \Delta V = K(L + \Delta L)^3. \quad (6a)$$

Now the modulus of elasticity Y is defined as the ratio of the change in stress to the change in strain, i.e.,

$$Y = \frac{\Delta\sigma}{(\Delta L/L)}. \quad (7a)$$

Therefore

$$V + \Delta V = KL^3 \left(1 + \frac{\Delta\sigma}{Y} \right)^3, \quad (8a)$$

so that

$$\Delta V = 3V \frac{\Delta\sigma}{Y}, \quad (9a)$$

neglecting the second- and third-order terms as $\Delta\sigma/Y$ is less than 10^{-2} .

From Eq. (16) we have

$$\Delta\sigma = CY\Delta p_s, \quad (10a)$$

where

$$C = \left(\frac{2.625L}{Yt} \right) \times 10^6. \quad (11a)$$

Substituting (10a) in (9a) gives ΔV in terms of Δp_s , viz.,

$$\Delta V = 3CV\Delta p_s. \quad (12a)$$

Eliminating ΔV from (5a) gives an expression for the increase in superpressure due to a forced change in height which takes into account the elastic stretching of the balloon film, namely,

$$\Delta p_s = -\frac{1}{6C} + \frac{\Delta p - p_I}{2} \pm \left[\frac{1}{36C^2} + \frac{\Delta p + p_I}{6C} + \frac{(\Delta p - p_I)^2}{4} \right]^{1/2}, \quad (13a)$$

and the increase in stress

$$\Delta\sigma = -\frac{Y}{6} + \frac{CY}{2}(\Delta p - p_I) \pm \left[\frac{Y^2}{36} + \frac{CY^2}{6}(\Delta p + p_I) + \frac{C^2Y^2}{4}(\Delta p - p_I)^2 \right]^{1/2}. \quad (14a)$$

Once Δp_s has been calculated from (13a), $\Delta\sigma$ may be calculated from (10a) and V from (12a).

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